AI / Machine Learning and Data Analytics

Q1. Which statement best describes Artificial Intelligence?
A) A system that performs tasks requiring human intelligence
B) A system used only for database management
C) A programming language for automation
D) A hardware architecture for computation
ANSWER: A

Q2. Machine Learning is best defined as:
A) Programming computers with fixed rules
B) Enabling systems to learn patterns from data
C) Designing microprocessors
D) Building computer networks
ANSWER: B

Q3. Which of the following is an example of Artificial Intelligence application?
A) Spam email filtering
B) Installing operating systems
C) Disk partitioning
D) File compression
ANSWER: A

Q4. Data Analytics primarily focuses on:
A) Extracting useful insights from data
B) Building network infrastructure
C) Managing operating systems
D) Designing processors
ANSWER: A

Q5. Which type of AI focuses on performing a specific task such as voice recognition?
A) Narrow AI
B) Strong AI
C) Artificial Super Intelligence
D) Symbolic AI
ANSWER: A

Q6. Which of the following best describes a dataset?
A) A collection of related data points
B) A network communication protocol
C) A database query language
D) A system software component
ANSWER: A

Q7. In machine learning, the data used to train a model is called:
A) Training data
B) Validation code
C) System data
D) Process data
ANSWER: A

Q8. Which term refers to the output predicted by a machine learning model?
A) Target variable
B) Kernel function
C) Hyperparameter
D) Activation unit
ANSWER: A

Q9. Which of the following best represents the role of data scientists?
A) Extract insights from data using statistical and computational methods
B) Manage hardware infrastructure
C) Install operating systems
D) Configure network devices
ANSWER: A

Q10. Which field combines statistics, programming, and domain knowledge to analyze data?
A) Data Science
B) Software Engineering
C) Network Engineering
D) Computer Architecture
ANSWER: A

Q11. Which concept measures how strongly two variables are related?
A) Correlation
B) Compilation
C) Hashing
D) Indexing
ANSWER: A

Q12. Which algorithm is widely used to minimize error during model training?
A) Gradient Descent
B) Binary Search
C) Breadth First Search
D) Depth First Search
ANSWER: A

Q13. Which metric measures the average squared difference between predicted and actual values?
A) Mean Squared Error
B) Precision
C) Recall
D) Accuracy
ANSWER: A

Q14. Which statistical measure represents the middle value of a sorted dataset?
A) Median
B) Mean
C) Variance
D) Standard deviation
ANSWER: A

Q15. Which concept describes how spread out the values of a dataset are?
A) Variance
B) Median
C) Mode
D) Frequency
ANSWER: A

Q16. Which measure represents the average of a dataset?
A) Mean
B) Mode
C) Range
D) Frequency
ANSWER: A

Q17. Which probability value indicates an impossible event?
A) 0
B) 1
C) 0.5
D) 10
ANSWER: A

Q18. Which concept is used to measure uncertainty in information theory?
A) Entropy
B) Median
C) Frequency
D) Standard deviation
ANSWER: A

Q19. Which function is commonly used as an activation function in neural networks?
A) ReLU
B) Merge
C) Compile
D) Sort
ANSWER: A

Q20. Which distribution is commonly used in statistics and machine learning?
A) Normal distribution
B) Uniform routing
C) Binary routing
D) Packet distribution
ANSWER: A

Q21. Which Python library is widely used for numerical computing?
A) NumPy
B) Flask
C) Django
D) Tkinter
ANSWER: A

Q22. Which Python library is used for data manipulation and analysis?
A) Pandas
B) Socket
C) Requests
D) PyGame
ANSWER: A

Q23. Which library provides machine learning algorithms in Python?
A) Scikit-learn
B) Kivy
C) Pyglet
D) PyTest
ANSWER: A

Q24. Which function loads CSV data into a Pandas DataFrame?
A) read_csv()
B) open_file()
C) load_data()
D) read_table()
ANSWER: A

Q25. Which Python library is commonly used for data visualization?
A) Matplotlib
B) Threading
C) Pickle
D) Asyncio
ANSWER: A

Q26. Which data structure in Python stores key-value pairs?
A) Dictionary
B) List
C) Tuple
D) Set
ANSWER: A

Q27. Which Python library is widely used for deep learning?
A) TensorFlow
B) BeautifulSoup
C) Tkinter
D) Paramiko
ANSWER: A

Q28. Which function displays the first few rows of a Pandas DataFrame?
A) head()
B) show()
C) preview()
D) top()
ANSWER: A

Q29. Which Python library is commonly used for advanced visualization?
A) Seaborn
B) Socket
C) Logging
D) Argparse
ANSWER: A

Q30. Which file format is commonly used to store tabular data for analysis?
A) CSV
B) EXE
C) DLL
D) BIN
ANSWER: A

Q31. Data preprocessing is mainly used to:
A) Improve data quality before analysis
B) Install database systems
C) Compile programs
D) Configure servers
ANSWER: A

Q32. Which technique replaces missing data with average values?
A) Mean imputation
B) Clustering
C) Tokenization
D) Regression
ANSWER: A

Q33. Which technique converts categorical variables into numerical values?
A) Encoding
B) Sampling
C) Aggregation
D) Tokenization
ANSWER: A

Q34. Which scaling technique transforms values between 0 and 1?
A) Min-Max scaling
B) Standard deviation scaling
C) Random scaling
D) Binary scaling
ANSWER: A

Q35. Which process removes duplicate entries from a dataset?
A) Deduplication
B) Augmentation
C) Encoding
D) Tokenization
ANSWER: A

Q36. Which step handles noisy or inconsistent data?
A) Data cleaning
B) Data encryption
C) Data compression
D) Data streaming
ANSWER: A

Q37. Which process reduces the size of datasets while maintaining essential information?
A) Dimensionality reduction
B) Data duplication
C) Data encryption
D) Data compilation
ANSWER: A

Q38. Which technique scales data to have mean 0 and standard deviation 1?
A) Standardization
B) Tokenization
C) Clustering
D) Partitioning
ANSWER: A

Q39. Which method randomly selects a subset of data from a larger dataset?
A) Sampling
B) Sorting
C) Encoding
D) Compression
ANSWER: A

Q40. Which method converts text into tokens for NLP preprocessing?
A) Tokenization
B) Encryption
C) Compression
D) Partitioning
ANSWER: A

Q41. Which visualization helps understand distribution of a variable?
A) Histogram
B) Tree diagram
C) Network graph
D) Flowchart
ANSWER: A

Q42. Which plot is used to identify relationships between two variables?
A) Scatter plot
B) Pie chart
C) Radar chart
D) Area chart
ANSWER: A

Q43. Which visualization is best for detecting outliers?
A) Box plot
B) Pie chart
C) Flowchart
D) Bar chart
ANSWER: A

Q44. Which statistical value represents the most frequent value in a dataset?
A) Mode
B) Mean
C) Median
D) Variance
ANSWER: A

Q45. Which plot shows proportions of categories in a dataset?
A) Pie chart
B) Histogram
C) Scatter plot
D) Box plot
ANSWER: A

Q46. Which plot shows relationships between variables using color intensity?
A) Heatmap
B) Pie chart
C) Histogram
D) Tree diagram
ANSWER: A

Q47. Which EDA step helps identify skewness in data?
A) Distribution analysis
B) Compilation
C) Encryption
D) Compression
ANSWER: A

Q48. Which measure indicates asymmetry in data distribution?
A) Skewness
B) Median
C) Mode
D) Range
ANSWER: A

Q49. Which measure describes peakedness of distribution?
A) Kurtosis
B) Correlation
C) Frequency
D) Precision
ANSWER: A

Q50. The main purpose of exploratory data analysis is to:
A) Understand patterns and structure in data
B) Install operating systems
C) Configure databases
D) Manage hardware
ANSWER: A

Q51. Which supervised learning task predicts a continuous numerical value?
A) Regression
B) Clustering
C) Association rule mining
D) Dimensionality reduction
ANSWER: A

Q52. Which algorithm constructs a tree-like structure for classification decisions?
A) Decision Tree
B) K-Means
C) Apriori
D) PCA
ANSWER: A

Q53. Which algorithm classifies data by finding the optimal hyperplane that separates classes?
A) Support Vector Machine
B) Linear Regression
C) K-Means
D) Apriori
ANSWER: A

Q54. Logistic Regression is primarily used for:
A) Classification problems
B) Clustering tasks
C) Dimensionality reduction
D) Data visualization
ANSWER: A

Q55. Which algorithm classifies data points based on the majority class of nearest neighbors?
A) K-Nearest Neighbors
B) Random Forest
C) Gradient Boosting
D) Naive Bayes
ANSWER: A

Q56. Which supervised learning algorithm assumes independence among predictors?
A) Naive Bayes
B) Decision Tree
C) Random Forest
D) K-Means
ANSWER: A

Q57. Which regression method fits a linear relationship between input variables and output?
A) Linear Regression
B) PCA
C) DBSCAN
D) Apriori
ANSWER: A

Q58. Which metric measures the proportion of correctly classified instances?
A) Accuracy
B) Entropy
C) Variance
D) Gradient
ANSWER: A

Q59. Precision measures:
A) Correct positive predictions among predicted positives
B) Correct predictions among all samples
C) Correct negative predictions only
D) Total dataset size
ANSWER: A

Q60. Recall measures:
A) Correct positive predictions among actual positives
B) Correct predictions among all classes
C) Correct negative predictions only
D) Dataset variance
ANSWER: A

Q61. Ensemble learning refers to:
A) Combining multiple models to improve prediction performance
B) Using a single powerful model
C) Increasing dataset size only
D) Removing noisy data
ANSWER: A

Q62. Which ensemble technique trains multiple models independently on different subsets of data?
A) Bagging
B) Boosting
C) Stacking
D) Sampling
ANSWER: A

Q63. Random Forest is primarily based on:
A) Bagging with decision trees
B) Boosting with neural networks
C) Clustering algorithms
D) Regression models only
ANSWER: A

Q64. Which boosting method assigns higher weights to misclassified samples?
A) AdaBoost
B) PCA
C) DBSCAN
D) Apriori
ANSWER: A

Q65. Gradient Boosting builds models:
A) Sequentially to reduce residual errors
B) Randomly without feedback
C) In parallel without dependencies
D) Only using unsupervised methods
ANSWER: A

Q66. Which ensemble technique combines predictions of several models using a meta-model?
A) Stacking
B) Bagging
C) Sampling
D) Scaling
ANSWER: A

Q67. The main advantage of ensemble methods is:
A) Improved predictive accuracy
B) Reduced dataset size
C) Faster compilation time
D) Smaller storage requirements
ANSWER: A

Q68. Which algorithm is widely used in machine learning competitions?
A) Gradient Boosting
B) K-Means
C) Naive Bayes
D) Linear Regression
ANSWER: A

Q69. XGBoost is an implementation of:
A) Gradient Boosting
B) Decision Trees
C) Logistic Regression
D) Clustering
ANSWER: A

Q70. Ensemble methods help reduce:
A) Overfitting and variance
B) Network latency
C) Disk fragmentation
D) CPU temperature
ANSWER: A

Q71. Which learning type uses unlabeled data?
A) Unsupervised Learning
B) Supervised Learning
C) Reinforcement Learning
D) Transfer Learning
ANSWER: A

Q72. Which algorithm groups similar data points into clusters?
A) K-Means
B) Linear Regression
C) Decision Tree
D) Random Forest
ANSWER: A

Q73. K-Means clustering works by:
A) Minimizing distance between points and cluster centroids
B) Maximizing classification accuracy
C) Building decision trees
D) Training neural networks
ANSWER: A

Q74. Which algorithm detects clusters based on density of data points?
A) DBSCAN
B) Logistic Regression
C) KNN
D) Linear Regression
ANSWER: A

Q75. Which technique reduces dimensionality while preserving variance?
A) Principal Component Analysis
B) Logistic Regression
C) Random Forest
D) Decision Tree
ANSWER: A

Q76. Which algorithm is commonly used for association rule mining?
A) Apriori
B) KNN
C) Linear Regression
D) SVM
ANSWER: A

Q77. Which concept measures how similar data points are in clustering?
A) Distance metric
B) Loss function
C) Activation function
D) Learning rate
ANSWER: A

Q78. Hierarchical clustering builds clusters:
A) In a tree-like hierarchy
B) Using neural networks
C) Using regression models
D) Randomly
ANSWER: A

Q79. Which technique reduces number of features in a dataset?
A) Dimensionality reduction
B) Tokenization
C) Encryption
D) Compression
ANSWER: A

Q80. Which distance metric is commonly used in clustering algorithms?
A) Euclidean distance
B) Binary distance
C) Routing distance
D) Packet distance
ANSWER: A

Q81. Which dataset is used to train a machine learning model?
A) Training dataset
B) Test dataset
C) Validation dataset
D) Feature dataset
ANSWER: A

Q82. Which dataset is used to evaluate final model performance?
A) Test dataset
B) Training dataset
C) Raw dataset
D) Feature dataset
ANSWER: A

Q83. Cross-validation helps:
A) Evaluate model performance on multiple subsets
B) Reduce dataset size
C) Remove features
D) Increase CPU speed
ANSWER: A

Q84. In k-fold cross validation, the dataset is divided into:
A) k subsets
B) two equal subsets
C) random clusters
D) hierarchical trees
ANSWER: A

Q85. Which curve shows trade-off between true positive rate and false positive rate?
A) ROC curve
B) Learning curve
C) Loss curve
D) Precision curve
ANSWER: A

Q86. Which metric combines precision and recall into a single score?
A) F1 Score
B) Accuracy
C) Entropy
D) Variance
ANSWER: A

Q87. Which matrix summarizes classification results?
A) Confusion Matrix
B) Distance Matrix
C) Feature Matrix
D) Kernel Matrix
ANSWER: A

Q88. Overfitting occurs when:
A) Model memorizes training data but fails on new data
B) Model performs poorly on training data
C) Model has fewer parameters
D) Dataset is too small
ANSWER: A

Q89. Underfitting occurs when:
A) Model is too simple to capture patterns in data
B) Model perfectly fits training data
C) Dataset has many features
D) Data is normalized
ANSWER: A

Q90. Which technique helps prevent overfitting?
A) Regularization
B) Encryption
C) Compilation
D) Compression
ANSWER: A

Q91. Feature engineering refers to:
A) Creating meaningful input variables from raw data
B) Installing machine learning software
C) Managing network traffic
D) Compressing datasets
ANSWER: A

Q92. Feature scaling is important because:
A) Some algorithms are sensitive to magnitude of features
B) It increases CPU speed
C) It reduces dataset size
D) It encrypts data
ANSWER: A

Q93. Which technique converts categorical variables into binary vectors?
A) One-hot encoding
B) Sampling
C) Clustering
D) Partitioning
ANSWER: A

Q94. Which method removes irrelevant features from datasets?
A) Feature selection
B) Tokenization
C) Compression
D) Encryption
ANSWER: A

Q95. Which method evaluates feature importance using decision trees?
A) Tree-based feature importance
B) PCA
C) K-Means
D) DBSCAN
ANSWER: A

Q96. Which transformation reduces skewness in numerical features?
A) Log transformation
B) Encoding
C) Sampling
D) Partitioning
ANSWER: A

Q97. Feature extraction differs from feature selection because it:
A) Creates new features from existing ones
B) Removes datasets entirely
C) Deletes observations
D) Encrypts variables
ANSWER: A

Q98. Which algorithm is commonly used for dimensionality reduction?
A) PCA
B) Naive Bayes
C) Random Forest
D) Decision Tree
ANSWER: A

Q99. Which feature selection method evaluates correlation between features and target variable?
A) Filter method
B) Sampling method
C) Compression method
D) Encryption method
ANSWER: A

Q100. Feature engineering improves models mainly by:
A) Increasing predictive power of features
B) Reducing memory usage only
C) Changing hardware architecture
D) Removing algorithms
ANSWER: A

Q101. A neural network consists of:
A) Layers of interconnected artificial neurons
B) Only a single processing unit
C) A group of database tables
D) A set of network routers
ANSWER: A

Q102. Which layer receives the initial input data in a neural network?
A) Input layer
B) Hidden layer
C) Output layer
D) Pooling layer
ANSWER: A

Q103. Which algorithm is used to update weights in neural networks during training?
A) Backpropagation
B) Binary Search
C) Depth First Search
D) Linear Search
ANSWER: A

Q104. Which activation function outputs values between 0 and 1?
A) Sigmoid
B) ReLU
C) Linear
D) Softplus
ANSWER: A

Q105. Which activation function helps reduce the vanishing gradient problem?
A) ReLU
B) Sigmoid
C) Tanh
D) Linear
ANSWER: A

Q106. The main purpose of activation functions in neural networks is to:
A) Introduce non-linearity into the model
B) Store data permanently
C) Increase storage capacity
D) Compress datasets
ANSWER: A

Q107. Which function converts outputs of a neural network into probabilities?
A) Softmax
B) ReLU
C) Tanh
D) Linear
ANSWER: A

Q108. The process of adjusting model weights based on error is called:
A) Training
B) Sampling
C) Clustering
D) Encoding
ANSWER: A

Q109. Which parameter controls how much the model weights change during training?
A) Learning rate
B) Batch size
C) Epoch number
D) Dataset size
ANSWER: A

Q110. Which technique helps prevent overfitting in neural networks?
A) Dropout
B) Tokenization
C) Encryption
D) Clustering
ANSWER: A

Q111. Which neural network architecture is designed for sequential data?
A) Recurrent Neural Network
B) Decision Tree
C) Random Forest
D) K-Means
ANSWER: A

Q112. Which neural network architecture is widely used for image processing tasks?
A) Convolutional Neural Network
B) Recurrent Neural Network
C) Logistic Regression
D) KNN
ANSWER: A

Q113. In convolutional neural networks, convolutional layers are mainly used to:
A) Extract spatial features from images
B) Store training data
C) Reduce dataset size
D) Increase CPU speed
ANSWER: A

Q114. Which operation reduces spatial dimensions in CNNs?
A) Pooling
B) Encoding
C) Tokenization
D) Sampling
ANSWER: A

Q115. Which neural network component stores learned parameters for feature detection in CNNs?
A) Filters
B) Tokens
C) Clusters
D) Trees
ANSWER: A

Q116. Long Short-Term Memory (LSTM) networks are designed to:
A) Capture long-term dependencies in sequence data
B) Classify images only
C) Compress datasets
D) Remove noise from data
ANSWER: A

Q117. Which architecture introduced the attention mechanism widely used in NLP models?
A) Transformer
B) CNN
C) Decision Tree
D) Logistic Regression
ANSWER: A

Q118. Attention mechanisms allow models to:
A) Focus on relevant parts of the input sequence
B) Reduce dataset size
C) Compress network weights
D) Remove training data
ANSWER: A

Q119. Which technique is commonly used to initialize neural network weights?
A) Xavier initialization
B) Random search
C) Linear scaling
D) Binary encoding
ANSWER: A

Q120. Batch normalization helps by:
A) Stabilizing and accelerating training
B) Compressing datasets
C) Reducing dataset size
D) Encrypting data
ANSWER: A

Q121. Natural Language Processing primarily deals with:
A) Processing and understanding human language
B) Image recognition
C) Network security
D) Hardware design
ANSWER: A

Q122. Tokenization refers to:
A) Splitting text into words or tokens
B) Encrypting text
C) Compressing text files
D) Translating text
ANSWER: A

Q123. Which representation technique converts words into dense vectors?
A) Word embeddings
B) Clustering
C) Encryption
D) Indexing
ANSWER: A

Q124. Which algorithm is widely used for generating word embeddings?
A) Word2Vec
B) K-Means
C) Apriori
D) Random Forest
ANSWER: A

Q125. Which task determines whether text expresses positive or negative sentiment?
A) Sentiment analysis
B) Topic modeling
C) Named entity recognition
D) Tokenization
ANSWER: A

Q126. Stop words are:
A) Common words removed during preprocessing
B) Rare words used for training
C) Words representing entities
D) Words used for classification
ANSWER: A

Q127. Which NLP task identifies names of people, places, or organizations in text?
A) Named Entity Recognition
B) Sentiment analysis
C) Tokenization
D) Clustering
ANSWER: A

Q128. Which model architecture powers modern NLP systems like BERT and GPT?
A) Transformer
B) CNN
C) Decision Tree
D) KNN
ANSWER: A

Q129. Stemming in NLP refers to:
A) Reducing words to their root form
B) Translating words
C) Encoding words
D) Sorting sentences
ANSWER: A

Q130. Which NLP technique discovers hidden themes in documents?
A) Topic modeling
B) Regression
C) Clustering
D) Encoding
ANSWER: A

Q131. Computer vision focuses on:
A) Enabling machines to interpret visual data
B) Managing databases
C) Designing processors
D) Encrypting network traffic
ANSWER: A

Q132. Which task identifies objects present in an image?
A) Object detection
B) Sentiment analysis
C) Tokenization
D) Regression
ANSWER: A

Q133. Which dataset is widely used for image classification research?
A) ImageNet
B) WikiText
C) LibriSpeech
D) CIFAR-Text
ANSWER: A

Q134. Image classification involves:
A) Assigning labels to images
B) Detecting network packets
C) Sorting numerical values
D) Compressing images
ANSWER: A

Q135. Which technique increases dataset diversity using transformations like rotation or flipping?
A) Data augmentation
B) Tokenization
C) Encryption
D) Sampling
ANSWER: A

Q136. Which type of neural network is most effective for image analysis?
A) Convolutional Neural Network
B) Recurrent Neural Network
C) Logistic Regression
D) Naive Bayes
ANSWER: A

Q137. Which operation detects edges or patterns in images in CNNs?
A) Convolution
B) Encoding
C) Tokenization
D) Sampling
ANSWER: A

Q138. Semantic segmentation assigns:
A) A class label to each pixel in an image
B) A label to the entire image only
C) Multiple labels to datasets
D) Clusters to images
ANSWER: A

Q139. Object detection differs from classification because it:
A) Identifies object location and class
B) Only assigns labels
C) Only detects edges
D) Only compresses images
ANSWER: A

Q140. Facial recognition systems are applications of:
A) Computer vision
B) Data compression
C) Network security
D) Compiler design
ANSWER: A

Q141. Big Data refers to datasets that are:
A) Extremely large and complex to process using traditional tools
B) Small and easy to process
C) Only stored in relational databases
D) Limited to numerical values
ANSWER: A

Q142. Which framework is widely used for distributed data processing?
A) Apache Spark
B) NumPy
C) Matplotlib
D) Pandas
ANSWER: A

Q143. Hadoop primarily provides:
A) Distributed storage and processing of large datasets
B) Local file compression
C) Database indexing
D) Image processing
ANSWER: A

Q144. Which component of Hadoop stores large datasets across distributed systems?
A) HDFS
B) SQL Server
C) FTP
D) TCP/IP
ANSWER: A

Q145. MapReduce works by:
A) Dividing tasks into map and reduce operations
B) Sorting datasets only
C) Compressing files
D) Encrypting data
ANSWER: A

Q146. Which characteristic of big data refers to the speed at which data is generated?
A) Velocity
B) Volume
C) Variety
D) Veracity
ANSWER: A

Q147. Which characteristic refers to the different types of data formats?
A) Variety
B) Velocity
C) Volume
D) Value
ANSWER: A

Q148. Which tool is commonly used for distributed big data storage?
A) Hadoop Distributed File System
B) SQLite
C) PostgreSQL
D) MySQL
ANSWER: A

Q149. Spark is preferred over Hadoop MapReduce mainly because:
A) It processes data in memory for faster computation
B) It uses less storage
C) It only processes images
D) It requires no programming
ANSWER: A

Q150. Which concept refers to extracting useful insights from large datasets?
A) Big Data Analytics
B) Data compression
C) Packet routing
D) File partitioning
ANSWER: A

Q151. Model deployment refers to:
A) Integrating a trained model into a production environment
B) Training a model from scratch
C) Collecting new datasets
D) Removing model parameters
ANSWER: A

Q152. Which technology is commonly used to package machine learning applications into containers?
A) Docker
B) NumPy
C) Pandas
D) OpenCV
ANSWER: A

Q153. MLOps focuses on:
A) Automating and managing machine learning lifecycle
B) Designing hardware circuits
C) Installing operating systems
D) Managing networks
ANSWER: A

Q154. Which tool is widely used for experiment tracking in machine learning?
A) MLflow
B) MySQL
C) Apache HTTP
D) FileZilla
ANSWER: A

Q155. Continuous integration in machine learning pipelines ensures:
A) Automated testing and integration of code changes
B) Larger dataset storage
C) Increased network bandwidth
D) Faster disk operations
ANSWER: A

Q156. Which concept ensures that deployed models maintain performance over time?
A) Model monitoring
B) Data compression
C) Tokenization
D) Encryption
ANSWER: A

Q157. Which API framework is commonly used to deploy machine learning models in Python?
A) FastAPI
B) Tkinter
C) Matplotlib
D) SciPy
ANSWER: A

Q158. Version control systems help by:
A) Tracking changes in code and models
B) Increasing CPU speed
C) Reducing dataset size
D) Encrypting datasets
ANSWER: A

Q159. Which platform is commonly used for managing ML workflows?
A) Kubeflow
B) MySQL
C) SQLite
D) Redis
ANSWER: A

Q160. Container orchestration systems like Kubernetes help in:
A) Managing large-scale model deployments
B) Compiling programs
C) Designing algorithms
D) Encrypting networks
ANSWER: A

Q161. Algorithmic bias occurs when:
A) AI systems produce unfair outcomes due to biased data or design
B) Models run faster
C) Networks fail
D) Data is encrypted
ANSWER: A

Q162. Data privacy refers to:
A) Protecting personal information from unauthorized access
B) Compressing datasets
C) Increasing storage capacity
D) Speeding up algorithms
ANSWER: A

Q163. Which principle ensures fairness and accountability in AI systems?
A) Ethical AI
B) Data compression
C) Network routing
D) Compiler optimization
ANSWER: A

Q164. Which type of attack manipulates input data to mislead AI models?
A) Adversarial attack
B) SQL injection
C) Packet flooding
D) Memory leak
ANSWER: A

Q165. Which regulation protects personal data of individuals in many countries?
A) GDPR
B) HTTP
C) TCP
D) SMTP
ANSWER: A

Q166. Transparency in AI refers to:
A) Making model decisions understandable to humans
B) Increasing data size
C) Removing training data
D) Compressing models
ANSWER: A

Q167. Explainable AI focuses on:
A) Interpreting how machine learning models make decisions
B) Encrypting neural networks
C) Compressing datasets
D) Increasing network bandwidth
ANSWER: A

Q168. Data anonymization helps by:
A) Removing identifiable personal information
B) Increasing dataset size
C) Encrypting networks
D) Improving GPU speed
ANSWER: A

Q169. Which concept ensures AI systems respect human rights and fairness?
A) Responsible AI
B) Binary computing
C) Packet routing
D) Compiler design
ANSWER: A

Q170. Privacy-preserving machine learning focuses on:
A) Protecting sensitive data during model training
B) Increasing model size
C) Compressing networks
D) Removing algorithms
ANSWER: A

Q171. If a model performs well on training data but poorly on unseen data, the problem is:
A) Overfitting
B) Underfitting
C) Regularization
D) Feature scaling
ANSWER: A

Q172. If a model is too simple to capture patterns in data, the problem is:
A) Underfitting
B) Overfitting
C) Cross-validation
D) Dimensionality reduction
ANSWER: A

Q173. Increasing training data size generally helps reduce:
A) Overfitting
B) Dataset variance
C) CPU speed
D) Disk fragmentation
ANSWER: A

Q174. Regularization techniques such as L1 and L2 are used to:
A) Prevent overfitting
B) Increase dataset size
C) Reduce GPU usage
D) Encrypt models
ANSWER: A

Q175. Feature scaling is especially important for algorithms based on:
A) Distance calculations
B) Decision trees
C) Rule-based systems
D) Database queries
ANSWER: A

Q176. Which evaluation metric is most useful when class distribution is highly imbalanced?
A) F1 Score
B) Accuracy
C) Variance
D) Mean
ANSWER: A

Q177. Hyperparameters differ from model parameters because they:
A) Are set before training begins
B) Are learned during training
C) Are always constant
D) Are dataset features
ANSWER: A

Q178. Increasing model complexity usually:
A) Increases risk of overfitting
B) Reduces training time
C) Eliminates noise
D) Removes features
ANSWER: A

Q179. Dimensionality reduction helps by:
A) Reducing number of input features while preserving information
B) Increasing dataset size
C) Encrypting data
D) Compiling algorithms
ANSWER: A

Q180. Ensemble learning improves prediction performance mainly because:
A) Multiple models reduce individual model errors
B) Datasets become smaller
C) Algorithms run slower
D) Networks become faster
ANSWER: A

Q181. A model trained on a dataset performs poorly because many features are highly correlated. Which technique can help address this issue?
A) Principal Component Analysis
B) Tokenization
C) Data duplication
D) Random sampling
ANSWER: A

Q182. Which technique divides data into training and testing subsets to evaluate a model's performance?
A) Train-test split
B) Tokenization
C) Data encryption
D) Data compression
ANSWER: A

Q183. Which method helps estimate the generalization ability of a model using multiple data splits?
A) Cross-validation
B) Clustering
C) Sampling
D) Encoding
ANSWER: A

Q184. If precision is high but recall is low, the model likely:
A) Misses many actual positive cases
B) Predicts too many positive cases
C) Performs perfectly
D) Has no classification ability
ANSWER: A

Q185. Which technique reduces the effect of large feature values dominating distance calculations?
A) Feature scaling
B) Data duplication
C) Encryption
D) Data partitioning
ANSWER: A

Q186. When the number of features exceeds the number of observations, the dataset may suffer from:
A) Curse of dimensionality
B) Network congestion

C) Data redundancy
D) Hardware failure
ANSWER: A

Q187. Which concept measures the uncertainty or impurity in decision tree learning?
A) Entropy
B) Variance
C) Accuracy
D) Gradient
ANSWER: A

Q188. Information gain in decision trees is used to:
A) Select the best feature for splitting
B) Increase dataset size
C) Reduce storage usage
D) Encrypt data
ANSWER: A

Q189. Increasing the depth of a decision tree generally leads to:
A) Higher risk of overfitting
B) Lower accuracy
C) Smaller datasets
D) Faster training
ANSWER: A

Q190. Which method combines predictions from multiple models by averaging their outputs?
A) Bagging
B) Encoding
C) Clustering
D) Sampling
ANSWER: A

Q191. Which metric is most appropriate when false negatives are very costly?
A) Recall
B) Accuracy
C) Precision
D) Variance
ANSWER: A

Q192. In binary classification, the true positive rate is also called:
A) Recall
B) Precision
C) Specificity

D) Accuracy
ANSWER: A

Q193. The ROC curve plots:
A) True positive rate vs false positive rate
B) Precision vs recall
C) Loss vs accuracy
D) Error vs variance
ANSWER: A

Q194. Which metric summarizes classifier performance across all classification thresholds?
A) AUC
B) Accuracy
C) Entropy
D) Variance
ANSWER: A

Q195. When a model performs equally poorly on training and test data, the issue is usually:
A) Underfitting
B) Overfitting
C) Data leakage
D) Sampling bias
ANSWER: A

Q196. Which technique introduces a penalty term to model complexity?
A) Regularization
B) Clustering
C) Tokenization
D) Compression
ANSWER: A

Q197. L1 regularization is also known as:
A) Lasso regression
B) Ridge regression
C) Logistic regression
D) Polynomial regression
ANSWER: A

Q198. L2 regularization is commonly called:
A) Ridge regression
B) Lasso regression
C) Linear regression

D) Polynomial regression
ANSWER: A

Q199. Early stopping during neural network training helps to:
A) Prevent overfitting
B) Increase dataset size
C) Reduce GPU usage
D) Encrypt model parameters
ANSWER: A

Q200. Batch size in neural network training refers to:
A) Number of samples processed before updating weights
B) Total dataset size
C) Number of features
D) Number of classes
ANSWER: A

Q201. Transfer learning involves:
A) Reusing knowledge from a pretrained model for a new task
B) Training models from scratch
C) Deleting datasets
D) Compressing neural networks
ANSWER: A

Q202. Which technique improves model robustness by adding variations to training data?
A) Data augmentation
B) Encoding
C) Compression
D) Clustering
ANSWER: A

Q203. Reinforcement learning involves:
A) Learning through rewards and penalties
B) Training on labeled data only
C) Grouping data into clusters
D) Compressing features
ANSWER: A

Q204. In reinforcement learning, the agent interacts with:
A) Environment
B) Dataset
C) Database

D) Compiler
ANSWER: A

Q205. The value function in reinforcement learning estimates:
A) Expected future reward
B) Dataset size
C) Model accuracy
D) Network latency
ANSWER: A

Q206. Which reinforcement learning algorithm updates policies directly?
A) Policy gradient methods
B) Decision trees
C) Logistic regression
D) K-Means
ANSWER: A

Q207. Exploration in reinforcement learning refers to:
A) Trying new actions to discover better rewards
B) Repeating the same action always
C) Removing features
D) Compressing datasets
ANSWER: A

Q208. Exploitation in reinforcement learning means:
A) Choosing the best-known action
B) Random action selection
C) Increasing dataset size
D) Compressing models
ANSWER: A

Q209. Deep reinforcement learning combines:
A) Reinforcement learning with deep neural networks
B) Clustering with regression
C) Encryption with compression
D) Sorting with indexing
ANSWER: A

Q210. AlphaGo is an example of:
A) Deep reinforcement learning system
B) Supervised regression model
C) Database system

D) Networking protocol
ANSWER: A

Q211. Generative models aim to:
A) Generate new data samples similar to training data
B) Compress datasets
C) Encrypt files
D) Sort data
ANSWER: A

Q212. Which model architecture generates realistic images using two competing networks?
A) Generative Adversarial Network
B) Decision Tree
C) Naive Bayes
D) Logistic Regression
ANSWER: A

Q213. GANs consist of:
A) Generator and discriminator networks
B) Encoder and decoder
C) Mapper and reducer
D) Predictor and classifier
ANSWER: A

Q214. Which model architecture learns compressed representations of data?
A) Autoencoder
B) Decision Tree
C) Linear Regression
D) KNN
ANSWER: A

Q215. Variational Autoencoders are used for:
A) Generative modeling
B) Clustering
C) Sorting
D) Encryption
ANSWER: A

Q216. Which concept describes models that adapt to new tasks using minimal data?
A) Few-shot learning
B) Batch learning
C) Online learning

D) Incremental learning
ANSWER: A

Q217. Which learning paradigm updates models continuously with incoming data?
A) Online learning
B) Batch learning
C) Transfer learning
D) Reinforcement learning
ANSWER: A

Q218. Federated learning allows models to be trained:
A) Across multiple devices without sharing raw data
B) Only on central servers
C) Without training data
D) Only on GPUs
ANSWER: A

Q219. Edge AI refers to:
A) Running AI models on local devices instead of cloud servers
B) Cloud-based AI systems only
C) Database systems
D) Networking systems
ANSWER: A

Q220. Explainable AI techniques aim to:
A) Interpret model predictions
B) Increase GPU memory
C) Compress neural networks
D) Encrypt datasets
ANSWER: A

Q221. Hyperparameter tuning aims to:
A) Find optimal configuration of model parameters
B) Increase dataset size
C) Encrypt model weights
D) Compress datasets
ANSWER: A

Q222. Grid search finds optimal hyperparameters by:
A) Testing all combinations in a defined search space
B) Randomly selecting values once
C) Removing parameters

D) Compressing models
ANSWER: A

Q223. Random search improves efficiency by:
A) Sampling hyperparameters randomly
B) Testing every possible value
C) Removing datasets
D) Encrypting features
ANSWER: A

Q224. Bayesian optimization selects hyperparameters based on:
A) Probabilistic modeling of performance
B) Random selection
C) Feature removal
D) Dataset duplication
ANSWER: A

Q225. Model interpretability becomes critical in domains such as:
A) Healthcare and finance
B) Image compression
C) File storage
D) Hardware design
ANSWER: A

Q226. Feature scaling is particularly important for algorithms that rely on:
A) Distance calculations
B) Rule-based systems
C) Decision trees only
D) Database queries
ANSWER: A

Q227. Which technique converts categorical variables into multiple binary columns?
A) One-hot encoding
B) Tokenization
C) Sampling
D) Partitioning
ANSWER: A

Q228. Which method selects the most important subset of features for model training?
A) Feature selection
B) Data augmentation
C) Data replication

D) Data compression
ANSWER: A

Q229. Wrapper methods for feature selection evaluate features by:
A) Training models with different feature subsets
B) Removing random features
C) Compressing datasets
D) Encrypting data
ANSWER: A

Q230. Embedded feature selection methods perform selection:
A) During model training
B) After model deployment
C) Before data preprocessing
D) After dataset deletion
ANSWER: A

Q231. When class distribution is highly imbalanced, which metric is usually more informative than accuracy?
A) F1 Score
B) Mean
C) Variance
D) Standard deviation
ANSWER: A

Q232. Precision-recall curves are particularly useful when:
A) The dataset is imbalanced
B) The dataset is small
C) The dataset is perfectly balanced
D) The dataset has no labels
ANSWER: A

Q233. Data leakage occurs when:
A) Information from test data influences training
B) Datasets are encrypted
C) Features are normalized
D) Data is duplicated
ANSWER: A

Q234. Which method helps estimate model performance when dataset size is small?
A) Cross-validation
B) Random deletion
C) Encryption

D) Compression
ANSWER: A

Q235. Stratified sampling ensures that:
A) Class distribution remains consistent across subsets
B) Dataset size increases
C) Features are removed
D) Models are compressed
ANSWER: A

Q236. Which optimizer adapts learning rates for each parameter individually?
A) Adam optimizer
B) Gradient descent
C) Linear regression
D) K-Means
ANSWER: A

Q237. Stochastic Gradient Descent updates weights:
A) Using one sample or small batches at a time
B) Using the entire dataset always
C) Without calculating gradients
D) Only after deployment
ANSWER: A

Q238. Vanishing gradient problem mainly affects:
A) Deep neural networks
B) Linear regression models
C) Decision trees
D) K-Means clustering
ANSWER: A

Q239. Gradient clipping is used to:
A) Prevent exploding gradients during training
B) Remove features
C) Encrypt data
D) Compress models
ANSWER: A

Q240. Which technique randomly disables neurons during training to reduce overfitting?
A) Dropout
B) Pooling
C) Convolution

D) Tokenization
ANSWER: A

Q241. Word embeddings represent words as:
A) Dense numerical vectors
B) Binary values only
C) Text labels
D) Random strings
ANSWER: A

Q242. BERT is primarily used for:
A) Language understanding tasks
B) Image segmentation
C) Network routing
D) Database indexing
ANSWER: A

Q243. Machine translation involves:
A) Converting text from one language to another
B) Compressing text
C) Encrypting documents
D) Sorting sentences
ANSWER: A

Q244. Which evaluation metric is commonly used for machine translation quality?
A) BLEU score
B) Accuracy
C) Variance
D) Precision
ANSWER: A

Q245. Attention mechanisms improve NLP models by:
A) Focusing on relevant parts of input sequences
B) Increasing dataset size
C) Removing features
D) Compressing models
ANSWER: A

Q246. Image segmentation differs from classification because it:
A) Assigns labels to individual pixels
B) Assigns one label to the whole image
C) Compresses images

D) Removes objects
ANSWER: A

Q247. Bounding boxes in object detection represent:
A) Locations of detected objects
B) Dataset size
C) Network latency
D) Image compression rate
ANSWER: A

Q248. Transfer learning in computer vision often uses:
A) Pretrained CNN models
B) Random feature generators
C) Decision trees only
D) K-Means clustering
ANSWER: A

Q249. YOLO is widely used for:
A) Real-time object detection
B) Sentiment analysis
C) Regression tasks
D) Text classification
ANSWER: A

Q250. Optical character recognition (OCR) involves:
A) Extracting text from images
B) Compressing image files
C) Encrypting images
D) Detecting edges only
ANSWER: A

Q251. Distributed computing improves big data processing by:
A) Dividing tasks across multiple machines
B) Compressing datasets
C) Encrypting files
D) Reducing dataset size
ANSWER: A

Q252. Which technology stores structured and unstructured big data efficiently?
A) NoSQL databases
B) Excel spreadsheets
C) Text editors

D) Compilers
ANSWER: A

Q253. Apache Kafka is mainly used for:
A) Real-time data streaming
B) Image classification
C) Regression modeling
D) Clustering datasets
ANSWER: A

Q254. Data lakes store:
A) Raw structured and unstructured data
B) Only relational tables
C) Only compressed data
D) Only images
ANSWER: A

Q255. Stream processing analyzes data:
A) In real-time as it is generated
B) After storage only
C) Only weekly
D) Only offline
ANSWER: A

Q256. Model versioning helps by:
A) Tracking different versions of trained models
B) Reducing dataset size
C) Encrypting algorithms
D) Compressing features
ANSWER: A

Q257. Continuous deployment ensures:
A) Automated release of models to production
B) Manual model updates only
C) Dataset encryption
D) Feature compression
ANSWER: A

Q258. A/B testing in machine learning deployment is used to:
A) Compare performance of two models in production
B) Encrypt models
C) Remove datasets

D) Compress models
ANSWER: A

Q259. Monitoring deployed models helps detect:
A) Model performance degradation
B) Network failures only
C) Disk errors
D) File compression errors
ANSWER: A

Q260. Data drift occurs when:
A) Data distribution changes over time
B) Dataset size increases
C) Data is encrypted
D) Models are compressed
ANSWER: A

Q261. Bias in AI systems often originates from:
A) Biased training data
B) GPU hardware
C) Network routers
D) File storage systems
ANSWER: A

Q262. Fairness in AI aims to:
A) Ensure models treat different groups equitably
B) Increase dataset size
C) Compress data
D) Encrypt features
ANSWER: A

Q263. Differential privacy protects data by:
A) Adding noise to sensitive information
B) Compressing datasets
C) Removing labels
D) Increasing feature size
ANSWER: A

Q264. Which attack attempts to extract training data from a model?
A) Model inversion attack
B) Packet attack
C) Sorting attack

D) Compression attack
ANSWER: A

Q265. Federated learning enhances privacy because:
A) Raw data remains on local devices
B) Data is publicly shared
C) Data is removed completely
D) Models are deleted
ANSWER: A

Q266. Increasing the number of training epochs too much may cause:
A) Overfitting
B) Underfitting
C) Data loss
D) Encryption failure
ANSWER: A

Q267. Batch normalization helps neural networks by:
A) Stabilizing training and improving convergence
B) Increasing dataset size
C) Compressing weights
D) Removing gradients
ANSWER: A

Q268. Which method visualizes high-dimensional data in two or three dimensions?
A) t-SNE
B) Decision Tree
C) Linear regression
D) Naive Bayes
ANSWER: A

Q269. PCA reduces dimensionality by:
A) Transforming features into orthogonal components
B) Removing all features
C) Compressing images
D) Encrypting data
ANSWER: A

Q270. Ensemble models improve robustness because they:
A) Combine predictions from multiple models
B) Remove training data
C) Compress datasets

D) Encrypt algorithms
ANSWER: A

Q271. Recommendation systems are widely used in:
A) E-commerce platforms
B) Operating systems
C) Network routers
D) Compilers
ANSWER: A

Q272. Autonomous vehicles rely heavily on:
A) Computer vision and deep learning
B) Database queries
C) File compression
D) Sorting algorithms
ANSWER: A

Q273. Chatbots primarily use:
A) Natural language processing
B) Image segmentation
C) Data compression
D) Database indexing
ANSWER: A

Q274. Fraud detection systems commonly apply:
A) Machine learning classification models
B) File compression
C) Sorting algorithms
D) Text editors
ANSWER: A

Q275. Predictive analytics aims to:
A) Forecast future trends using historical data
B) Compress data files
C) Encrypt networks
D) Remove datasets
ANSWER: A

Q276. Which learning paradigm uses rewards to guide learning?
A) Reinforcement learning
B) Supervised learning
C) Unsupervised learning

D) Transfer learning
ANSWER: A

Q277. Which technique reduces model complexity by removing unnecessary parameters?
A) Pruning
B) Encoding
C) Tokenization
D) Compression
ANSWER: A

Q278. Edge computing processes data:
A) Near the data source
B) Only in cloud servers
C) Only in data centers
D) Only in databases
ANSWER: A

Q279. Model interpretability is critical in fields such as:
A) Healthcare diagnostics
B) Image compression
C) File storage
D) Hardware design
ANSWER: A

Q280. Which concept ensures AI systems remain accountable and transparent?
A) Responsible AI
B) Binary computing
C) Packet routing
D) Compiler design
ANSWER: A

Q281. Which process automatically selects best hyperparameters during training?
A) Hyperparameter tuning
B) Tokenization
C) Encoding
D) Sampling
ANSWER: A

Q282. Which algorithm groups similar documents based on topic similarity?
A) Clustering
B) Regression
C) Classification

D) Encryption
ANSWER: A

Q283. Which technique reduces noise in datasets by smoothing values?
A) Data smoothing
B) Data duplication
C) Data encryption
D) Data deletion
ANSWER: A

Q284. Which type of learning uses labeled examples during training?
A) Supervised learning
B) Reinforcement learning
C) Unsupervised learning
D) Transfer learning
ANSWER: A

Q285. Which model is commonly used for time-series forecasting?
A) Recurrent Neural Network
B) Decision Tree
C) K-Means
D) Naive Bayes
ANSWER: A

Q286. Which metric evaluates prediction error in regression models?
A) Mean Absolute Error
B) Precision
C) Recall
D) F1 Score
ANSWER: A

Q287. Which technique reduces training time by using parallel computations on GPUs?
A) Hardware acceleration
B) Tokenization
C) Sampling
D) Encoding
ANSWER: A

Q288. Which concept measures information content in a variable?
A) Entropy
B) Accuracy
C) Precision

D) Recall
ANSWER: A

Q289. Which architecture processes sequential text efficiently?
A) Transformer
B) Decision Tree
C) Logistic Regression
D) KNN
ANSWER: A

Q290. Which technique helps models generalize better by adding small noise to training data?
A) Data augmentation
B) Data deletion
C) Data encryption
D) Data partitioning
ANSWER: A

Q291. Which algorithm builds multiple decision trees and aggregates their outputs?
A) Random Forest
B) K-Means
C) Apriori
D) PCA
ANSWER: A

Q292. Which evaluation method repeatedly trains models on different dataset splits?
A) Cross-validation
B) Sampling
C) Tokenization
D) Encoding
ANSWER: A

Q293. Which method reduces high-dimensional data while preserving variance?
A) PCA
B) Decision Tree
C) KNN
D) Naive Bayes
ANSWER: A

Q294. Which concept ensures secure handling of sensitive user data?
A) Data privacy
B) Data compression
C) Data sorting

D) Data partitioning
ANSWER: A

Q295. Which technique helps detect anomalies in data?
A) Anomaly detection
B) Regression
C) Tokenization
D) Encoding
ANSWER: A

Q296. Which AI field focuses on analyzing images and videos?
A) Computer vision
B) NLP
C) Reinforcement learning
D) Data mining
ANSWER: A

Q297. Which AI field focuses on understanding human language?
A) Natural language processing
B) Computer vision
C) Reinforcement learning
D) Data compression
ANSWER: A

Q298. Which platform allows scalable training of ML models using distributed clusters?
A) Apache Spark
B) NumPy
C) Tkinter
D) OpenCV
ANSWER: A

Q299. Which principle ensures AI systems do not harm individuals or society?
A) Ethical AI
B) Binary logic
C) File compression
D) Packet routing
ANSWER: A

Q300. The primary goal of artificial intelligence is to:
A) Develop systems capable of performing intelligent tasks
B) Replace all software systems
C) Compress all datasets

D) Encrypt computer networks
ANSWER: A